

# Topological Spaces

## **General Topology**

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## Definition

Given a set  $X$ , a **topology** on  $X$  is a collection  $\mathcal{T}$  of subsets of  $X$ , called the **open subsets of  $X$** , satisfying the following conditions:

Top 1)  $X$  and  $\emptyset$  are elements of  $\mathcal{T}$ .

Top 2)  $\mathcal{T}$  is closed under finite intersections.

Top 3)  $\mathcal{T}$  is closed under arbitrary unions.

The pair  $(X, \mathcal{T})$ , where  $\mathcal{T}$  is a topology on  $X$ , is called a **topological space**. The elements of the underlying space  $X$  are usually referred to as its **points**. When both  $X$  and its topology are understood, we will simply refer to the elements in  $\mathcal{T}$  as **open subsets**.

## Definition

If  $X$  is a topological space and  $p \in X$ , an **open neighbourhood** (or simply, a **neighbourhood**) of  $p$  is an open subset of  $X$  containing  $p$ . More generally, if  $K \subseteq X$ , a **neighbourhood of the subset  $K$**  is an open subset containing  $K$ .

## Definition

Given two topologies  $\mathcal{T}_1$  and  $\mathcal{T}_2$  on  $X$ , we say that  **$\mathcal{T}_1$  is finer than  $\mathcal{T}_2$**  if  $\mathcal{T}_1 \supseteq \mathcal{T}_2$ . In this case, we also say that  **$\mathcal{T}_2$  is coarser than  $\mathcal{T}_1$** .

## Example 2.1

Let  $X$  be any set whatsoever, and let  $\mathcal{T} = \mathcal{P}(X)$ , so every subset of  $X$  is open. This is called the **discrete topology on  $X$** , and  $(X, \mathcal{T})$  is called a **discrete space**.

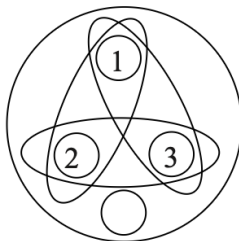


Figure 1: Discrete topology of  $\{1, 2, 3\}$

## Example 2.2

Let  $\mathcal{T} = \{Y, \emptyset\}$  for any set  $Y$ . This is called the **trivial topology** on  $Y$ .

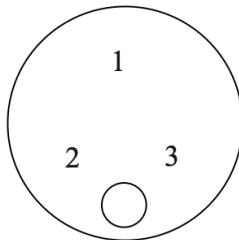


Figure 2: Trivial topology of  $\{1, 2, 3\}$

### Example 2.3

Let  $Z = \{1, 2, 3\}$ , and declare  $\mathcal{T} = \{\{1\}, \{1, 2\}, \{1, 2, 3\}, \emptyset\}$ , show that  $(Z, \mathcal{T})$  is a topological space.

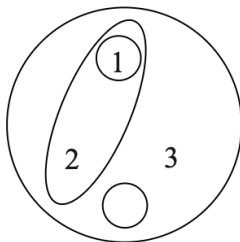


Figure 3: The topology  $\mathcal{T}$  on  $Z$

### Example 2.4

Show that  $(\mathbb{C}, \mathcal{T})$  is a topological space, where  $\mathcal{T}$  is defined as the family of subsets  $\Omega$  with the property that for every  $\omega \in \Omega$  there exists  $\mathbb{R} \ni \varepsilon > 0$  such that the set

$$B_\varepsilon(\omega) := \{z \in \mathbb{C} : |z - \omega| < \varepsilon\}$$

is contained in  $\Omega$ . This particular family  $\mathcal{T}$  is often referred to as the **standard topology** on  $\mathbb{C}$ .

### Example 2.5

*More generally, let  $(M, d)$  be a metric space, and let  $\mathcal{T}$  be the collection of subsets containing an open ball*

$$B_\varepsilon(x) := \{y \in M : d(y, x) < \varepsilon\}$$

*around each of its points. It can be shown that  $\mathcal{T}$  is a topology, called the **metric topology on  $M$** . If  $d$  is the Euclidean metric, the metric topology is called the **Euclidean topology**.*

## Example 2.6

*Here are some standard subsets of Euclidean spaces that we will use throughout the course. Unless otherwise specified, each of these is considered as a topological space with the Euclidean (metric) topology.*

- The **unit interval** is the subset  $I \subseteq \mathbb{R}$  defined by

$$I = [0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}.$$

- For any nonnegative integer  $n$ , the **(open) unit ball of dimension  $n$**  is the subset  $\mathbb{B}^n \subseteq \mathbb{R}^n$  consisting of all vectors of length strictly less than 1:

$$\mathbb{B}^n = \{x \in \mathbb{R}^n : |x| < 1\}.$$

Similarly, the subset  $\overline{\mathbb{B}}^n \subseteq \mathbb{R}^n$

$$\overline{\mathbb{B}}^n = \{x \in \mathbb{R}^n : |x| \leq 1\}.$$

We sometimes call  $\overline{\mathbb{B}}^2$  the **closed unit disk**.

- The **(unit) circle** is the subset  $S^1 \subseteq \mathbb{R}^2$  consisting of unit vectors in the plane:

$$S^1 = \{x \in \mathbb{R}^2 : |x| = 1\}.$$

It is often useful to identify the plane  $\mathbb{R}^2$  with the set  $\mathbb{C}$  of complex numbers by the correspondence  $(x, y) \leftrightarrow x + iy$ , and think of the circle as the set of complex numbers with unit modulus:

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

We will use whichever representation is most convenient for the problem at hand.

- The **(unit)  $n$ -sphere** is the subset  $S^n \subseteq \mathbb{R}^{n+1}$  consisting of unit vectors in  $\mathbb{R}^{n+1}$ :

$$S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}.$$

## Definition

*Since every subset of a Euclidean space is a metric space in its own right (with the restriction of the Euclidean metric), it automatically inherits a metric topology, called the **Euclidean topology**, and unless otherwise specified, subsets of  $\mathbb{R}^n$  are always considered as topological spaces with this topology.*

### Remark

*If  $X$  is a topological space and  $V$  is any open subset of  $X$ , then we can define a topology on  $V$  just by declaring the open subsets of  $V$  to be those open subsets of  $X$  that are contained in  $V$ .*

## Definition

*If  $X$  is a topological space, a subset  $F \subseteq X$  is said to be a **closed subset of  $X$**  if its complement  $X \setminus F$  is an open subset.*

## Definition

Let  $X$  be a topological space and let  $A$  be any subset of  $X$ . The **closure of  $A$  in  $X$** , denoted by  $\overline{A}$ , is the set

$$\overline{A} = \bigcap \{B \subseteq X : B \supseteq A \text{ and } B \text{ is closed in } X\}.$$

## Definition

Let  $X$  be a topological space and let  $A$  be any subset of  $X$ . The **interior** of  $A$ , denoted by  $\text{Int}(A)$ , is

$$\text{Int}(A) = \bigcup \{U \subseteq X : U \subseteq A \text{ and } U \text{ is open in } X\}.$$

We also define the **exterior** of  $A$ , denoted by  $\text{Ext}(A)$ , as  $\text{Ext}(A) = X \setminus \overline{A}$ , and the **boundary** of  $A$ , denoted by  $\partial A$ , as  $\partial A = X \setminus (\text{Int}(A) \cup \text{Ext}(A))$ .

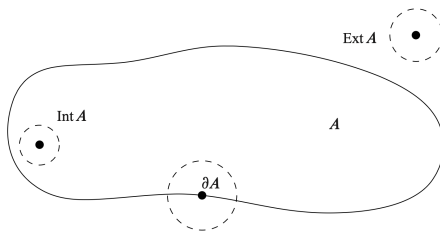


Figure 4: Interior, exterior, and boundary points

## Remark

*It follows from the definition that for any subset  $A \subseteq X$ ,*

$$X = \text{Int } A \bigsqcup \text{Ext } A \bigsqcup \partial A.$$

## Theorem 2.7

Let  $X$  be a topological space and let  $A \subseteq X$  be any subset.

- i) A point is in  $\text{Int}(A)$  if and only if it has a neighbourhood contained in  $A$ .
- ii) A point is in  $\text{Ext}(A)$  if and only if it has a neighbourhood contained in  $X \setminus A$ .
- iii) A point is in  $\partial A$  if and only if every neighbourhood of it contains both a point of  $A$  and a point of  $X \setminus A$ .
- iv) A point is in  $\overline{A}$  if and only if every neighbourhood of it contains a point of  $A$ .
- v)  $\overline{A} = A \cup \partial A = \text{Int}(A) \cup \partial A$ .
- vi)  $\text{Int}(A)$  and  $\text{Ext}(A)$  are open in  $X$ , while  $\overline{A}$  and  $\partial A$  are closed in  $X$ .

vii) *The following are equivalent:*

- a)  *$A$  is open in  $X$ .*
- b)  *$A = \text{Int}(A)$ .*
- c)  *$A$  contains none of its boundary points.*
- d) *Every point of  $A$  has a neighbourhood contained in  $A$ .*

viii) *The following are equivalent:*

- a')  *$A$  is closed in  $X$ .*
- b')  *$A = \overline{A}$ .*
- c')  *$A$  contains all of its boundary points.*
- d') *Every point of  $X \setminus A$  has a neighbourhood contained in  $X \setminus A$ .*

## Definition

Given a topological space  $X$  and a set  $A \subseteq X$ , we say that a point  $p \in X$  is a **limit point of  $A$**  if every neighbourhood of  $p$  contains a point of  $A$  other than  $p$ . Limit points are also sometimes called **accumulation points** or **cluster points**. A point  $p \in A$  is called an **isolated point of  $A$**  if  $p$  has a neighbourhood  $U$  in  $X$  such that  $U \cap A = \{p\}$ .

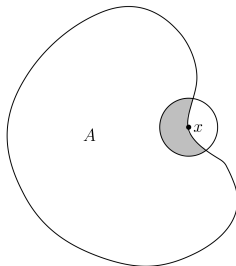


Figure 5: Limit point of a set

## Definition

A **sequence** in a topological space  $(X, \mathcal{T})$  is a map  $S : \mathbb{N} \rightarrow X$ . If  $S$  is a sequence, we will usually denote the value of  $S$  at  $n$  by the symbol  $x_n$  rather than using the function notation  $S(n)$ . The values  $x_n$  are also called the **terms** or the **elements** of the sequence. The notation  $(x_n)$  is commonly used to denote this sequence.

## Definition

Let  $X$  be a topological space and let  $x \in X$ . We say that a sequence  $(x_n)$  in  $X$  **converges to  $x$** , and  **$x$  is a limit of the sequence**, if for every neighbourhood  $U$  of  $x$  there exists  $N \in \mathbb{N}$  such that  $x_n \in U$  for all  $n \geq N$ . Symbolically, this is denoted by  $x_n \rightarrow x$  or  $\lim x_n = x$ .

## Definition

A function  $f : X \rightarrow Y$  between topological spaces is said to be **continuous** if

$$f^{-1}(\mathcal{T}_Y) \subseteq \mathcal{T}_X.$$

## Theorem 2.8

*A map between topological spaces is continuous if and only if the preimage of every closed subset is closed.*

## Proposition 2.9

*Let  $X, Y$ , and  $Z$  be topological spaces.*

- i) Every constant map  $f : X \rightarrow Y$  is continuous.*
- ii) The identity map  $\text{Id}_X : X \rightarrow X$  is continuous.*
- iii) If  $f : X \rightarrow Y$  is continuous, so is the restriction of  $f$  to any open subset of  $X$ .*
- iv) If  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  are both continuous, then so is their composition  $g \circ f : X \rightarrow Z$ .*

## Theorem 2.10 (Local Criterion for Continuity)

*A map  $f : X \rightarrow Y$  between topological spaces is continuous if and only if each point  $x \in X$  has a neighbourhood on which (the restriction of)  $f$  is continuous.*

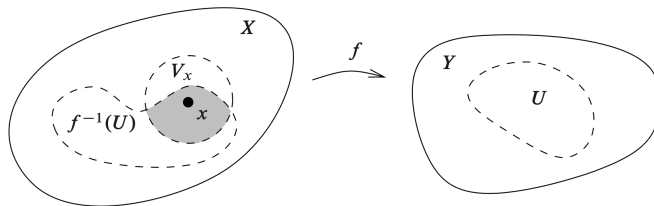


Figure 6: Local criterion for continuity

## Definition

If  $f$  is a bijection and its inverse  $f^{-1} : Y \rightarrow X$  is continuous, the function is called a **homeomorphism** and the topological spaces  $X$  and  $Y$  are said to be **homeomorphic** (or **topologically equivalent**), written  $X \approx Y$ . The family of homeomorphisms  $f : X \rightarrow Y$  is denoted by  $\text{Homeo}(X, Y)$ . Further, we say that  $f$  is a **local homeomorphism** if every point  $x \in X$  has a neighbourhood  $U \subseteq X$  such that  $f(U)$  is an open subset of  $Y$  and  $f|_U \in \text{Homeo}(U, f(U))$ .

### Example 2.11

*Any open ball in  $\mathbb{R}^n$  is homeomorphic to any other open ball; the homeomorphism can easily be constructed as a composition of **translations**  $x \mapsto x + x_0$  and **dilations**  $x \mapsto cx$ . Similarly, all spheres in  $\mathbb{R}^n$  are homeomorphic to each other. These examples illustrate that “size” is not a topological property.*

## Definition

A map  $f : X \rightarrow Y$  (continuous or not) is said to be an **open map** if it takes open subsets of  $X$  to open subsets of  $Y$ , and it is said to be a **closed map** if it takes closed subsets of  $X$  to closed subsets of  $Y$ .

## Theorem 2.12

*If  $f : X \rightarrow Y$  is a continuous bijective map, then  $f$  is a homeomorphism if and only if  $f$  is open.*

### Theorem 2.13 (Properties of Local Homeomorphisms)

- a) *Every homeomorphism is a local homeomorphism.*
- b) *Every local homeomorphism is continuous and open.*
- c) *Every bijective local homeomorphism is a homeomorphism.*

### Theorem 2.14

*Let  $Y$  be a topological space equipped with the trivial topology. Then every sequence in  $Y$  converges to every point of  $Y$ .*

## Definition

A topological space is said to be **Hausdorff** (or to satisfy a **Hausdorff condition**) if for every pair of points  $x, y \in X$  there exist open neighbourhoods  $U$  of  $x$  and  $V$  of  $y$  such that  $U \cap V = \emptyset$ . A topological space satisfying such property is known as a **Hausdorff space**.

## Theorem 2.15

*Every metric space  $(M, d)$  is Hausdorff.*

## Theorem 2.16

*Let  $X$  be a Hausdorff space.*

- i) Every finite subset of  $X$  is closed.*
- ii) If a sequence  $(p_n)$  in  $X$  converges to a limit  $p \in X$ , then the limit is unique.*

## Definition

Let  $(X, \mathcal{T})$  be a topological space. A collection  $\mathcal{B}$  of subsets of  $X$  is called a **basis for the topology of  $X$**  if the following two conditions hold:

- i) Every element of  $\mathcal{B}$  is an open subset of  $X$ .
- ii) Every open subset of  $X$  is the union of some collection of elements of  $\mathcal{B}$ .

If the topology is understood, we will just say that  $\mathcal{B}$  is a **basis for  $X$** .

## Theorem 2.17

*Suppose  $X$  is a topological space, and  $\mathcal{B}$  is a basis for its topology. Then a subset  $U \subseteq X$  is open if and only if it satisfies the **basis criterion** with respect to  $\mathcal{B}$ , i.e.*

*for each  $p \in U$ , there exists  $B \in \mathcal{B}$  such that  $p \in B \subseteq U$ .*

*Conversely, let  $\mathcal{C}$  be a collection of open subsets on  $X$  such that for each open set  $U$  of  $X$  and each  $x \in U$ , there is an element  $C$  of  $\mathcal{C}$  such that  $x \in C \subseteq U$ . Then  $\mathcal{C}$  is a basis for the topology of  $X$ .*

### Theorem 2.18

*The collection of all open balls in  $\mathbb{R}^n$  is a basis for its Euclidean topology. Furthermore, the collection*

$$\mathcal{B} = \{B_r(x) : r \text{ is rational and } x \text{ has rational coordinates}\}$$

*is also a basis.*

### Theorem 2.19

*Let  $X$  and  $Y$  be topological spaces and let  $\mathcal{B}$  be a basis for  $Y$ . A map  $f : X \rightarrow Y$  is continuous if and only if for every basis subset  $B \in \mathcal{B}$ , the subset  $f^{-1}(B)$  is open in  $X$ . Similarly, a map  $g : Y \rightarrow X$  is open if and only if for every basis subset  $B \in \mathcal{B}$ , the subset  $g(B)$  is open in  $X$ .*

### Remark

*Not every collection of sets can be a basis for a topology. The next theorem gives necessary and sufficient conditions for a collection of subsets of  $X$  to be a basis for some topology on  $X$ .*

## Theorem 2.20

*Let  $X$  be a set, and suppose  $\mathcal{B}$  is a collection of subsets of  $X$ . Then  $\mathcal{B}$  is a basis for some topology on  $X$  if and only if it satisfies the following two conditions:*

- i)  $\bigcup_{B \in \mathcal{B}} B = X$ .
- ii) *If  $B_1, B_2 \in \mathcal{B}$  and  $x \in B_1 \cap B_2$ , there exists an element  $B_3 \in \mathcal{B}$  such that  $x \in B_3 \subseteq B_1 \cap B_2$ .*

*In this case, there is a unique topology on  $X$  for which  $\mathcal{B}$  is a basis, namely, the collection of all unions of elements of  $\mathcal{B}$*

$$\mathcal{T} = \left\{ \bigcup \mathcal{C} \mid \mathcal{C} \subseteq \mathcal{B} \right\}.$$

*This unique topology  $\mathcal{T}$  is called the **topology generated by  $\mathcal{B}$** .*

## Remark

*Whereas the Hausdorff property ensures that a topological space has “enough” open subsets to confirm our spatial intuition, for many purposes (including the study of manifolds), it is useful to restrict our attention to spaces that do not have “too many” open subsets. It would be too restrictive to ask for countably many open subsets, because even  $\mathbb{R}$  has uncountably many. But it turns out that the right balance is struck by requiring the existence of a basis that is countable.*

## Definition

If  $X$  is a topological space and  $p \in X$ , a collection  $\mathcal{B}_p$  of neighbourhoods of  $p$  is called a **neighbourhood basis for  $X$  at  $p$**  if every neighbourhood of  $p$  contains some  $B \in \mathcal{B}_p$ . We say that  $X$  is **first countable** if there exists a countable neighbourhood basis at each point of  $X$ .

### Example 2.21

*If  $X$  is a metric space and  $p \in X$ , the set of balls  $B_r(p)$  with rational radii is a neighbourhood basis at  $p$ , so every metric space is first countable.*

## Definition

A topological space is said to be **second countable** if it admits a countable basis for its topology.

## Definition

If  $X$  is any topological space, a collection  $\mathcal{U}$  of subsets of  $X$  is said to be a **cover of  $X$**  (or to **cover  $X$** ) if every point in  $X$  is in at least one of the sets in  $\mathcal{U}$ . It is called an **open cover** [respectively, **closed cover**] if each of the sets in  $\mathcal{U}$  is open [respectively, closed]. Given any cover  $\mathcal{U}$ , a **subcover of  $\mathcal{U}$**  is a subcollection  $\mathcal{U}' \subseteq \mathcal{U}$  that still covers  $X$ .

## Definition

A topological space  $X$  is said to be **separable** if it contains a countable dense subset, and to be a **Lindelöf space** if every open cover of  $X$  has a countable subcover.

## Theorem 2.22 (Properties of Second Countable Spaces)

*Suppose  $X$  is a second countable space. Then*

- i)  *$X$  is first countable*
- ii)  *$X$  is separable.*
- iii)  *$X$  is a Lindelöf space.*

### Example 2.23

*Every Euclidean space is second countable.*

## Definition

A topological space  $M$  is said to be **locally Euclidean of dimension  $n$**  if every point of  $M$  has a neighbourhood in  $M$  that is homeomorphic to an open subset of  $\mathbb{R}^n$ .

### Theorem 2.24

*A topological space  $M$  is locally Euclidean of dimension  $n$  if and only if either of the following properties holds:*

- a) Every point of  $M$  has a neighbourhood homeomorphic to an open ball in  $\mathbb{R}^n$ .*
- b) Every point of  $M$  has a neighbourhood homeomorphic to  $\mathbb{R}^n$ .*

## Definition

Suppose  $M$  is locally Euclidean of dimension  $n$ . If  $U \subseteq M$  is an open subset that is homeomorphic to an open subset of  $\mathbb{R}^n$ , then  $U$  is called a **coordinate domain**, and any homeomorphism  $\varphi$  from  $U$  to an open subset of  $\mathbb{R}^n$  is called a **coordinate map**. The pair  $(U, \varphi)$  is called a **coordinate chart** (or just a **chart**) for  $M$ . A coordinate domain that is homeomorphic to a ball in  $\mathbb{R}^n$  is called a **coordinate ball**. (When  $M$  has dimension 2, we sometimes use the term **coordinate disk**.) The preceding lemma shows that every point in a locally Euclidean space is contained in a coordinate ball. If  $p \in M$  and  $U$  is a coordinate domain containing  $p$ , we say that  $U$  is a **coordinate neighbourhood** or **Euclidean neighbourhood** of  $p$ .

## Remark

Any chart  $(U, \varphi)$  associated to  $p$  induces a set of **coordinate functions**  $(x^1, x^2, \dots, x^n)$  defined by

$$x^\mu = \pi^\mu \circ \phi, \quad \mu = 1, \dots, n$$

where  $\pi^\mu$  is the  $\mu$ th-canonical projection on  $\mathbb{R}^n$ , and this implies that one can write  $\phi(p) = (x^1(p), x^2(p), \dots, x^n(p))$ .

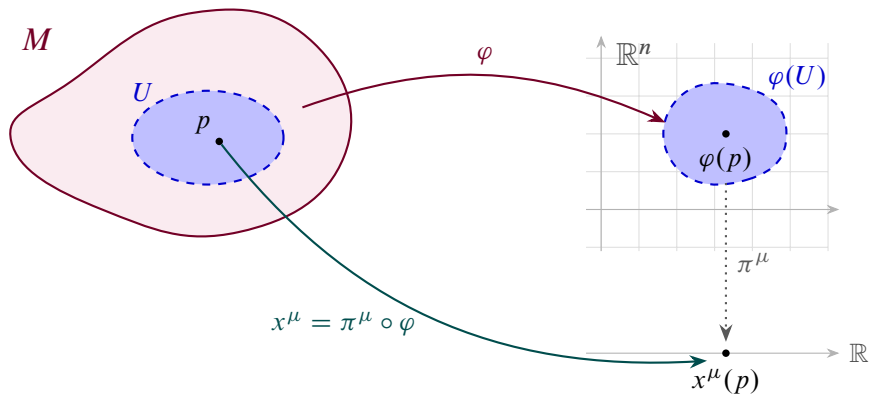


Figure 7: Chart on a point  $p$  in a locally Euclidean space  $M$

## Definition

An  **$n$ -dimensional topological manifold** is a second countable Hausdorff space that is locally Euclidean of dimension  $n$ .

## Theorem 2.25

*Every open subset of an  $n$ -manifold is an  $n$ -manifold.*

### Theorem 2.26 (Invariance of Dimension)

*If  $m \neq n$ , a nonempty topological space cannot be both an  $m$ -manifold and an  $n$ -manifold.*

## Theorem 2.27

*A separable metric space that is locally Euclidean of dimension  $n$  is an  $n$  manifold. Conversely, every manifold is separable and metrisable.*

## Definition

The **closed  $n$ -dimensional upper half space**  $\mathbb{H}^n$  is defined by

$$\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n : x^n \geq 0\}.$$

## Definition

An  **$n$ -dimensional manifold with boundary** is a second countable Hausdorff space in which every point has a neighbourhood homeomorphic either to an open subset of  $\mathbb{R}^n$ , or to an open subset of  $\mathbb{H}^n$ , considering  $\mathbb{H}^n$  as a topological space with the Euclidean topology.

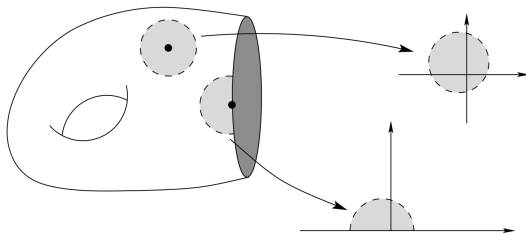


Figure 8: A manifold with boundary

## Definition

If  $M$  is an  $n$ -manifold with boundary, we define a **coordinate map for  $M$**  to be a pair  $(U, \varphi)$ , where  $U \subseteq M$  is open and  $\varphi$  is a homeomorphism from  $U$  to an open subset of either  $\mathbb{R}^n$  or  $\mathbb{H}^n$ . Just as in the case of manifolds, the set  $U$  is called a **coordinate domain**, and  $\varphi$  is called a **coordinate map**. When is important to make the distinction, we say that  $(U, \varphi)$  is an **interior chart** if  $\varphi(U)$  is an open subset of  $\mathbb{R}^n$  (which includes the case in which  $\varphi(U)$  is an open subset of  $\mathbb{H}^n$  that is contained in  $\text{Int } \mathbb{H}^n$ ); and a **boundary chart** if  $\varphi(U)$  is an open subset of  $\mathbb{H}^n$  with  $\varphi(U) \cap \partial \mathbb{H}^n \neq \emptyset$ . If  $M$  is an  $n$ -manifold with boundary, a point  $p \in M$  is called an **interior point of  $M$**  if it is in the domain of an interior chart; and it is called a **boundary point of  $M$**  if it is in the domain of a boundary chart that takes  $p$  to  $\partial \mathbb{H}^n$ . The **boundary of  $M$**  (or **manifold boundary**, if distinction is necessary), denoted by  $\partial M$ , is the set of all its boundary points, and its **interior**, denoted by  $\text{Int } M$ , is the set of all its interior points.

### Theorem 2.28

*If  $M$  is an  $n$ -dimensional manifold with boundary, then  $\text{Int } M$  is an open subset of  $M$ , which is itself an  $n$ -dimensional manifold without boundary.*

### Theorem 2.29 (Invariance of the Boundary)

*If  $M$  is a manifold with boundary, then a point of  $M$  cannot be both a boundary point and an interior point. Thus  $\partial M$  and  $\text{Int } M$  are disjoint subsets whose union is  $M$ .*

## Corollary

*If  $M$  is a nonempty  $n$ -dimensional manifold with boundary, then  $\partial M$  is closed in  $M$ , and  $M$  is an  $n$ -manifold if and only if  $\partial M = \emptyset$ .*